

# Bone Stacking Combinatorial Number System

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## Abstract

This paper defines a finite number system by enumerating combinatorial structures built from at most twelve abstract bones. A structure is first classified by its total number of bones, then by its structural partition, and finally by the orientations of its one-bone, two-bone, and three-bone piles. The resulting system contains exactly 2,911 distinct values. We derive the counting formula, give ranking and unranking procedures, and provide both concrete and generalized examples.

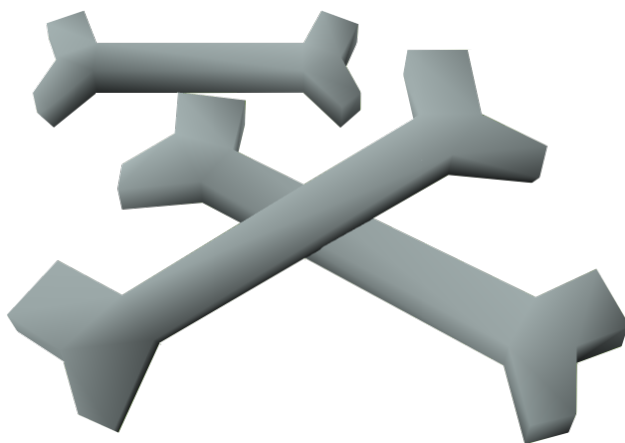


Figure 1: Bone number C1, representing decimal 9.

# 1 Introduction

The system was inspired by the observation that many small mammals have twelve pairs of ribs, as well as a curiosity surrounding the invention of roman numerals. The mathematical construction, however, treats the bones as abstract combinatorial objects. Up to twelve bones are used to represent a number, and are interpreted in a base-2911 number system to allow for individual digits to be read regardless of the positioning of the bones.

The central idea is to enumerate every distinct structure in a fixed order over three levels:

1. Structures are grouped by total number of bones: all one-bone structures, then all two-bone structures, and so on through twelve bones. This is to minimize the number of bones needed for small numbers, while expanding the total number of representable numbers.
2. Within a fixed bone-count, structures are grouped by their structural partition. For example, three bones can be partitioned as

$$1 + 1 + 1, \quad 2 + 1, \quad 3.$$

Because the piles are unordered,  $1 + 2$  is not a separate partition from  $2 + 1$ .

3. Within a fixed partition, the possible orientations are enumerated according to a fixed canonical ordering.

This gives a one-to-one correspondence between valid bone structures and the natural numbers.

$$1, 2, \dots, 2911.$$

## 2 Definitions

**Definition 2.1** (Pile). *A pile is a connected group of one, two, or three bones. Four bones is likely to topple over and is not considered. Bone piles can be represented by the alphabet: 1,A,B,C,D,E,F,G,H,I.*

**Definition 2.2** (Bone orientation). *A one-bone pile has one orientation of bone stackings:*

$$1 = \text{No connections.}$$

*A two-bone pile has three orientations:*

$$\begin{aligned} A &= \text{end-to-end,} \\ B &= \text{end-to-middle,} \\ C &= \text{middle-to-middle.} \end{aligned}$$

*A three-bone pile has six orientations:*

$$\begin{aligned} D &= AA = \text{end-to-end} + \text{end-to-end,} \\ E &= AB = \text{end-to-end} + \text{end-to-middle,} \\ F &= AC = \text{end-to-end} + \text{middle-to-middle,} \\ G &= BB = \text{end-to-middle} + \text{end-to-middle,} \\ H &= BC = \text{end-to-middle} + \text{middle-to-middle,} \\ I &= CC = \text{middle-to-middle} + \text{middle-to-middle.} \end{aligned}$$

*The ordering convention is*

$$A < B < C, \quad D < E < F < G < H < I.$$

**Definition 2.3** (Structural partition). *Suppose a structure consists of  $a$  one-bone piles,  $b$  two-bone piles, and  $c$  three-bone piles, where  $a, b, c \in \mathbb{N}_0$  and  $a + b + c \leq 12$ . The structural partition of a structure containing  $n$  bones is the triple  $(a, b, c)$  satisfying*

$$a + 2b + 3c = n.$$

*The piles are treated as unordered within each size class. Thus, for example, ABC represents the same structure regardless of the order in which the three two-bone piles are listed. The canonical representation is obtained by sorting orientations according to*

$$A < B < C$$

*for two-bone piles and*

$$D < E < F < G < H < I$$

*for three-bone piles.*

*When different pile sizes occur, three-bone piles are written first, followed by two-bone piles, followed by one-bone piles. Within a partition, smaller letters are written first.*

**Definition 2.4** (Bone-number). *A bone-number is the canonical symbolic representation of a valid structure. The decimal value of a bone-number is its position in the global enumeration described above.*

### 3 The Beginning of the Enumeration

The first few values are

$$\begin{array}{ll}
 1 = 1, & 9 = C1, \\
 2 = 11, & 10 = D, \\
 3 = A, & 11 = E, \\
 4 = B, & 12 = F, \\
 5 = C, & 13 = G, \\
 6 = 111, & 14 = H, \\
 7 = A1, & 15 = I, \\
 8 = B1, &
 \end{array}$$

Continuing,

$$\begin{array}{ll}
 16 = 1111, & 24 = BC1, \\
 17 = A11, & 25 = CC1, \\
 18 = B11, & 26 = D1, \\
 19 = C11, & 27 = E1, \\
 20 = AA1, & 28 = F1, \\
 21 = AB1, & 29 = G1, \\
 22 = AC1, & 30 = H1, \\
 23 = BB1, & 31 = I1.
 \end{array}$$

At the upper end of the system,

$$\begin{array}{l}
 2907 = HHHH, \\
 2908 = HHHI, \\
 2909 = HHII, \\
 2910 = HIII, \\
 2911 = IIII.
 \end{array}$$

Thus the largest value is

$$\boxed{2911},$$

## 4 Counting the Structures

The counting can be reduced to a simple multiset problem.

**Lemma 4.1.** *For a structural partition containing  $b$  two-bone piles, the number of possible orientation multisets is*

$$\binom{b+2}{2}.$$

*Proof.* There are three possible two-bone orientations,  $A, B, C$ . We must choose a multiset of size  $b$  from these three types. Equivalently, let  $x_A, x_B, x_C \geq 0$  satisfy

$$x_A + x_B + x_C = b.$$

By stars and bars, the number of solutions is

$$\binom{b+3-1}{3-1} = \binom{b+2}{2}.$$

□

**Lemma 4.2.** *For a structural partition containing  $c$  three-bone piles, the number of possible orientation multisets is*

$$\binom{c+5}{5}.$$

*Proof.* There are six possible three-bone orientations,

$$D, E, F, G, H, I.$$

A multiset of size  $c$  is determined by nonnegative integers

$$x_D + x_E + x_F + x_G + x_H + x_I = c.$$

Again by stars and bars, the number of solutions is

$$\binom{c+6-1}{6-1} = \binom{c+5}{5}.$$

□

**Proposition 4.3** (Structures in a fixed partition). *If a partition contains " $a$ " one-bone piles, " $b$ " two-bone piles, and " $c$ " three-bone piles, then the number of distinct structures is*

$$\boxed{\binom{b+2}{2} \binom{c+5}{5}}.$$

*Proof.* The one-bone piles have only one possible orientation, since all one-bone piles are identical. The two-bone piles contribute  $\binom{b+2}{2}$  orientations, and the three-bone piles contribute  $\binom{c+5}{5}$  orientations. These choices are independent, so the total is their product. □

## 5 Counting an Entire Bone Block

Let  $B_n$  denote the number of structures containing exactly  $n$  bones.

For each pair  $(b, c)$  satisfying

$$2b + 3c \leq n,$$

where  $b$  is the number of 2-bone structures and  $c$  is the number of 3-bone structures, the remaining number of bones is

$$a = n - 2b - 3c.$$

We first fix the number  $c$  of 3-bone structures and consider every possible value of  $c$ , from 0 up to  $\lfloor n/3 \rfloor$  since each multiple of 3-bone structures uses exactly 3 bones from the total  $n$ . For each fixed  $c$ , we then consider every possible number  $b$  of 2-bone structures that can be formed from the remaining bones, from 0 up to  $\lfloor (n - 3c)/2 \rfloor$ .

For each pair  $(b, c)$ , there are

$$\binom{b+2}{2}$$

ways to distribute the  $b$  2-bone structures among their three possible types, and

$$\binom{c+5}{5}$$

ways to distribute the  $c$  3-bone structures among their six possible types.

Therefore,

$$B_n = \sum_{c=0}^{\lfloor n/3 \rfloor} \sum_{b=0}^{\lfloor (n-3c)/2 \rfloor} \binom{b+2}{2} \binom{c+5}{5}.$$

Denote the cumulative number of structures through  $n$  bones.

$$C_n = \sum_{k=1}^n B_k$$

**Theorem 5.1** (Finite enumeration). *The construction defines exactly 2911 distinct bone-numbers.*

*Proof.* Every valid structure is uniquely defined by a total bone count  $n \in \{1, \dots, 12\}$ , a structural partition  $(a, b, c)$ , and a canonical multiset of orientations within each pile size. The enumeration orders these disjoint categories without overlap. Hence each structure occurs exactly once.

The number of structures in the  $n$ -bone block is  $B_n$ , so the total number of structures is

$$\sum_{n=1}^{12} B_n = C_{12} = 2911.$$

□

## 6 Ranking and Unranking

The conversion between decimal integers and bone-numbers can be understood as a pair of inverse operations.

### 6.1 Decimal to Bone-Number

Let  $N \in \{1, \dots, 2911\}$ . First find the unique  $n$  satisfying

$$C_{n-1} < N \leq C_n,$$

where  $C_0 = 0$ . The number lies in the  $n$ -bone block. Its one-indexed position within that block is

$$r = N - C_{n-1}.$$

Next enumerate the structural partitions of  $n$  bones. For each partition, add

$$\binom{b+2}{2} \binom{c+5}{5}$$

until the partition containing position  $r$  is found. If the cumulative number before the selected partition is  $P$ , then its one-indexed position within that partition is

$$s = r - P.$$

Finally, unrank the orientation multisets within the selected partition. In a partition containing  $b$  two-bone piles and  $c$  three-bone piles, define

$$M_2(b) = \binom{b+2}{2}, \quad M_3(c) = \binom{c+5}{5}.$$

The partition therefore contains

$$M_3(c)M_2(b)$$

structures. Structures are ordered by first enumerating the  $c$ -multiset of three-bone orientations, then the  $b$ -multiset of two-bone orientations, then the trivial  $a$ -multiset one-bone orientation. The one-indexed ranks of these two components are

$$p = \left\lfloor \frac{s-1}{M_2(b)} \right\rfloor + 1$$

and

$$q = (s-1) \bmod M_2(b) + 1.$$

Thus  $p$  identifies the  $p$ -th three-bone orientation multiset, while  $q$  identifies the  $q$ -th two-bone orientation multiset.

## 6.2 Decimal to Bone-Number Example: 2500

Consider the decimal number

$$2500.$$

Since

$$1847 < 2500 \leq 2911,$$

the number lies in the twelve-bone block. Its position inside that block is

$$2500 - 1847 = 653.$$

The relevant partition table begins

| Partition         | Number of structures | Cumulative |
|-------------------|----------------------|------------|
| $1^{12}$          | 1                    | 1          |
| $2 + 1^{10}$      | 3                    | 4          |
| $2^2 + 1^8$       | 6                    | 10         |
| $2^3 + 1^6$       | 10                   | 20         |
| $2^4 + 1^4$       | 15                   | 35         |
| $2^5 + 1^2$       | 21                   | 56         |
| $2^6$             | 28                   | 84         |
| $3 + 1^9$         | 6                    | 90         |
| $3 + 2 + 1^7$     | 18                   | 108        |
| $3 + 2^2 + 1^5$   | 36                   | 144        |
| $3 + 2^3 + 1^3$   | 60                   | 204        |
| $3 + 2^4 + 1$     | 90                   | 294        |
| $3^2 + 1^6$       | 21                   | 315        |
| $3^2 + 2 + 1^4$   | 63                   | 378        |
| $3^2 + 2^2 + 1^2$ | 126                  | 504        |
| $3^2 + 2^3$       | 210                  | 714        |

Because

$$504 < 653 \leq 714,$$

the required partition is

$$3^2 + 2^3.$$

Its local rank is

$$653 - 504 = 149.$$

Each choice of the two three-bone orientations is followed by ten choices of the three two-bone orientations. Hence

$$g = \left\lfloor \frac{149 - 1}{10} \right\rfloor + 1 = 15$$

and

$$q = (149 - 1) \bmod 10 + 1 = 9.$$

The 15-th three-bone orientation is

$$FI,$$

while the 9-th two-bone orientation is

$$BCC.$$

Therefore

$$\boxed{2500 = FIBCC}.$$

### 6.3 Bone-Number to Decimal

The reverse process is a ranking operation. Given a bone-number, its corresponding decimal value is determined by finding its position in the complete enumeration of structures.

For a general bone-number, proceed as follows:

1. Count the total number of bones  $n$ .
2. Determine its structural partition  $(a, b, c)$ , where  $a$ ,  $b$ , and  $c$  denote the numbers of one-, two-, and three-bone piles, respectively.
3. Add the complete blocks containing fewer than  $n$  bones. If  $C_{n-1}$  denotes the cumulative number of structures through  $n - 1$  bones, this contributes  $C_{n-1}$ .
4. Within the  $n$ -bone block, enumerate the structural partitions in the prescribed order and add the sizes of all partitions preceding the selected partition. Let this cumulative number be  $P$ .
5. Determine the one-indexed rank  $s$  of the given orientation multiset within the selected partition.

If  $P$  is the number of structures preceding the selected partition and  $s$  is the one-indexed rank within that partition, then the corresponding decimal value is

$$\boxed{N = C_{n-1} + P + s.}$$

### 6.4 Bone-Number to Decimal Example: "IX"

See Figure 1 for an example illustration.

Let the observed bone placements  $I$  and  $X$  denote the one-, two-bone orientations respectively:

$$I = 1,$$

$$X = C.$$

Since bone placement permutations are considered equivalent, we use the canonical representation

$$IX = C1.$$

This has structural partition

$$2 + 1,$$

Which contains three bones. The structures with fewer than three bones are

$$11, A, B, C.$$

Thus the cumulative number of structures through two bones is

$$C_2 = 5.$$

Within the three-bone block, the structural partitions preceding  $2 + 1$  consist only of

$$1 + 1 + 1.$$

This partition contains exactly one structure:

$$111.$$

Therefore,

$$P = 1.$$

Within the  $2 + 1$  partition, the possible two-bone orientations are

$$A1, B1, C1.$$

Thus  $C1$  is the third structure in this partition, so

$$s = 3.$$

Therefore the global rank is

$$N = C_2 + P + s = 5 + 1 + 3 = 9.$$

Hence

$$\boxed{IX'' = C1 = 9}.$$

## 7 A Flappy Bird Example

### 7.1 Encoding Game Stat and Program

The system can also be used as an abstract representation for the state of a simple game such as *Flappy Bird*. For example, one could assign bone-numbers to quantities such as

1.  $X$ -coordinate,
2.  $Y$ -coordinate,
3. velocity,
4. gravity,
5. flap impulse,
6. pipe  $X$ -positions,
7. pipe gap  $Y$ -positions,
8. score,
9. game speed,
10. random-number state.

Each state variable could be encoded using one or more values in the range  $1, \dots, 2911$ .

A deliberately conservative estimate of roughly 190 instructions and 10 states would require

$$200 \log_2(2911) \approx 2301$$

bits, or

$$\frac{2301}{8} \approx 288 \text{ bytes.}$$

This estimate represents the amount of memory needed to store game states and program instructions while excluding graphics, input/output, and other machine states. Since small mammals are imagined to provide two racks of ribs, this would correspond to 100 small mammals.

## 7.2 Encoding a $20 \times 20$ Monochrome Image

A  $20 \times 20$  monochrome image contains

$$20 \cdot 20 = 400$$

binary pixels, and therefore has

$$2^{400}$$

possible states.

Since one bone-number has 2911 possible values, the minimum number  $m$  of bone-numbers required satisfies

$$2911^m \geq 2^{400}.$$

Taking logarithms gives

$$m \geq \frac{400}{\log_2(2911)} \approx 34.76.$$

Hence

$$\boxed{m = 35}.$$

So 35 bone-numbers are sufficient to encode every possible  $20 \times 20$  monochrome image. Since each small mammal is imagined to provide two racks of ribs, this would correspond to 18 small mammals

## 8 Conclusion

The proposed number system is a finite combinatorial enumeration. A value is identified by the hierarchy

$$\boxed{\text{total bone count} \longrightarrow \text{structural partition} \longrightarrow \text{orientation multiset.}}$$

The number of structures for a fixed partition follows from multisets and stars and bars, giving

$$\binom{b+2}{2} \binom{c+5}{5}.$$

Summing over all partitions with at most twelve bones produces exactly 2911 distinct values.

The system therefore provides a well-defined bijection between the natural numbers  $1, \dots, 2911$  and a finite family of canonical bone structures. Decimal integers can be *unranked* into bone-numbers by locating their block, partition, and orientation rank; bone-numbers can be *ranked* back into integers by reversing the same sequence.

## 9 Take Away

Given a set of 12 ribs from a small mammal, you could arrange the bones in piles of 1, 2, and 3, dispersed over a small region, such that you represent the numbers from 1 to 2911. This alphabet of 2911 numerical values is made of only 10 primitive symbols, limited structurally, with equivalent permutations.